NetSuite Script Types

1. Differentiate Client side and Server-Side scripting. Mention at least 3 points for each with an example.

Ans.

	Client-Side Scripting	Server-Side Scripting
1	Client-side scripting runs in the user's web browser (client-side). It's executed on the user's device after the web page has been loaded.	Server-side scripting runs on NetSuite servers. It's executed before the web page is sent to the user's browser.
2	Client-side scripting is often used to enhance the user interface and create interactive features within the browser. For example, we can use it to perform client-side form validation, create pop-up dialogs, alerts or manipulate the appearance of the page without requiring a full server request.	Server-side scripting is used for more complex business logic, data processing, and interactions with the NetSuite database. It's commonly used for automating workflows, creating custom records and fields, and implementing business logic.
3	Client-side scripts can enhance the user experience by reducing the need for full page refreshes, but they may not be suitable for all tasks, especially those requiring server data or security-sensitive operations.	Server-side scripting is more secure than client-side scripting because it's executed on the server and can control access to sensitive data and perform server-side validation.
4	Client-side scripts always prefer simple UI manipulation and basic functionality. If complex operation is executed on client-side, then user page might get slow.	Server-side scripting is preferred for complex functionality. We can execute main logic at Server-side and then only result can render to Client-side script through APIs.
5	Example, Google Map – At client side, user can only view best route to reach his destination.	Example, Google Map – At server side, Dijkstra's Shortest Path Algorithm works to find shortest route.

2. Give one use case for each of the following script types
    a. Client Script
    b. User Event Script
    c. Suitelet
    d. Map-Reduce Script
    e. Scheduled Script
    f. Restlet
    g. Portlet

Ans.

AtoZ Shopping portal: Let suppose we have a shopping portal where we can shop for anything from groceries to electronic items. Then following scripts can be use in such a way:

Client Scripts:

- We can use client scripts to enhance the user interface for the customer.
- When user login/signup to portal that time client script can be used to validate his credential.
- This can also be used for notification alert for suggestions and offer/sale.
- Client script is basically used for UI design.
- We can use client script to categories our products like grocery, cosmetics, medicine etc.

User Event Scripts:

- We can event scripts to automate order processing.
- When a customer places an order on your website, a user event script can trigger automatic updates to inventory levels, generate purchase orders for suppliers, and update the financial records.
- We can use before load event to check whether user login to the portal or not, it has not login then before to load any categories of product we will redirect to login page
- Before Submit can use if user placing any order, then we will give alert to enter phone number and address if he has not entered because to ship order it's mandatory to have all these details
- After submit can use to redirect user to payment page after placing order

Suitelet Scripts:

- This can be used to create custom web forms for users or suppliers/shipment office to interact. We can create a suitelet to allow users to check their order status and shipment offices to update shipping status of order or supplier to update their product listings.
- Using suitlet we can make our shopping portal responsive and user friendly.
- We can add filters to our product categories as per price and branding

Maps reduce Scripts:

- This script can use to calculate available stock
- As we have large data of user and products then we can generate sales and marketing report from map reduce
- We can check in which cities we have the highest sales, which categories/products have the highest scale, etc. This will also help us to suggest products to the user.

Scheduled Scripts:

- We can use schedule script to mail user once order has been out to delivery
- We can also use schedule script to generate daily or weekly sales/offer notification to user

Restlet Scripts:

- We can use this script to integrate third party applications on portals like payment gate ways, UPI payment, Shipping carrier's API etc.

Portlet Scripts:

- Using this script, we can allow the user to add any product in Wishlist.
- We can allow user to customize his page as per need, only favorable categories item will be visible on user account.

3. Differentiate Map-Reduce and Scheduled script. Mention at least 5 points.

Ans.

	Map-Reduce script	Scheduled script
1	These scripts are typically used for processing large amounts of data in a parallel and efficient manner. They can be triggered manually or in response to a specific event, such as when a record is created or updated.	Scheduled scripts are designed to run on a predetermined schedule, such as daily, weekly, or monthly. They are commonly used for routine, repetitive tasks.
2	Map-Reduce scripts can handle large datasets and distribute the work among multiple server instances. This makes them suitable for optimizing performance when dealing with extensive data processing tasks.	Scheduled scripts run sequentially on a single server instance; they are not as efficient for handling large datasets as Map-Reduce scripts.
3	These are ideal for data transformation, data imports/exports, and complex data processing tasks. They are often used when you need to perform operations on a large number of records.	Scheduled scripts are commonly used for tasks like sending email reminders, generating reports, and performing routine maintenance operations.
4	Developing Map-Reduce scripts can be more complex as they require the implementation of both the map and reduce phases.	Scheduled scripts tend to be more straightforward to develop, making them a good choice for simple automation tasks.
5	Map-Reduce scripts offer more extensive logging capabilities and robust error handling to help you track and troubleshoot data processing.	Scheduled scripts also support logging and error handling but may not provide the same level of detail as Map-Reduce scripts.